

THE L^1 -GRADIENT ENDPOINT FOR BADLY APPROXIMABLE KRONECKER SEQUENCES

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STATUS

This is a candidate full answer, subject to human review, to both questions following Theorem 7 of Brown and Steinerberger [1]: the regular-grid L^1 -gradient estimate does hold uniformly for every badly approximable Kronecker sequence, so such an infinite sequence exists.

SOURCE QUESTION

For $d \geq 2$ and a badly approximable vector $\alpha \in \mathbb{R}^d$, put $x_j = j\alpha \pmod{1}$ in $\mathbb{T}^d$. The source proves, uniformly in N , that

$$\left| \int_{\mathbb{T}^d} f - \frac{1}{N} \sum_{j=1}^N f(x_j) \right| \lesssim_{\alpha, d} N^{-1/d} \|\nabla f\|_{\infty}^{(d-1)/d} \|\nabla f\|_2^{1/d}.$$

For a regular grid it proves the stronger endpoint with $\|\nabla f\|_1$ in place of $\|\nabla f\|_2$, and asks whether the same is true for the Kronecker sequence. More generally, it asks whether any single infinite sequence attains that endpoint uniformly in N .

in terms of L^p -spaces that seems to be new.

Theorem 7. *We have, for some explicit constant c_d depending only on the dimension, for all differentiable $f : [0, 1]^d \rightarrow \mathbb{R}$ sampled on the regular grid $(x_k)_{k=1}^N$*

$$\left| \int_{[0,1]^d} f(x) dx - \frac{1}{N} \sum_{k=1}^N f(x_k) \right| \leq c_d \|\nabla f\|_{L^\infty(\mathbb{T}^d)}^{(d-1)/d} \|\nabla f\|_{L^1(\mathbb{T}^d)}^{1/d} N^{-1/d}.$$

We observe that this is a slightly better estimate than Theorem 6 (an $L^1(\mathbb{T}^d)$ norm instead of the larger $L^2(\mathbb{T}^d)$ norm); this is maybe to be expected since one would assume that stronger estimates become available for the regular grid. We will also show that this is the best possible bound in terms of these L^p -spaces. It is an interesting question whether this bound (L^1 instead of L^2) is also true for the Kronecker sequence (Theorem 6). More generally, one could ask whether there is a sequence $(x_n)_{n=1}^\infty$ that uniformly attains the same error estimate as Theorem 7.

2.5. Other manifolds. Nothing about our approach is particularly tied to the

PROOF INTUITION

The source splits the empirical measure into a short heat-flow part and a smoothed part. It bounds the first only by transportation distance, forcing an L^∞ gradient, and bounds the second in $\dot{H}^{-1}$, forcing an L^2 gradient. Keep the same split, but represent both pieces as divergences of vector fields.

The low-frequency field has L^2 norm $O(N^{-1/d})$ by the source's own Fourier estimate. The high-frequency field is a sum of truncated Coulomb kernels. Bad approximability separates the first N orbit points at scale $N^{-1/d}$; a packing argument then puts this sum in the critical weak space $L^{d/(d-1),\infty}$ with the same norm. Lorentz Hölder duality against $L^{d,1}$, followed by the elementary interpolation

$$\|g\|_{L^{d,1}} \lesssim_d \|g\|_\infty^{(d-1)/d} \|g\|_1^{1/d},$$

gives exactly the missing endpoint.

CRITICAL HEAT-FLOW FIELD

Write $h = N^{-1/d}$ and

$$\mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{x_j}.$$

Let $P_t = e^{t\Delta}$ be heat flow on $\mathbb{T}^d$, with kernel p_t , and take $t = h^2$. We use Lorentz spaces with respect to normalized Lebesgue measure.

Lemma 1 (Separated heat atoms). *Suppose $x_1, \dots, x_N \in \mathbb{T}^d$ satisfy*

$$\min_{i \neq j} d(x_i, x_j) \geq ah.$$

There is a vector field F_{hi} such that

$$\operatorname{div} F_{\text{hi}} = \mu_N - P_t \mu_N \quad \text{and} \quad \|F_{\text{hi}}\|_{L^{d/(d-1),\infty}} \leq C_{a,d} h.$$

Proof. In distributions, set

$$F_{\text{hi}}(x) = -\frac{1}{N} \sum_{j=1}^N K_t(x - x_j), \quad K_t(z) = \int_0^t \nabla p_s(z) ds.$$

Since $\partial_s p_s = \Delta p_s$, taking the divergence gives the asserted identity. The standard Gaussian gradient estimate for the torus heat kernel implies, with $r = d(z, 0)$,

$$(1) \quad |K_t(z)| \leq C_d r^{1-d} \exp\left(-\frac{r^2}{C_d t}\right).$$

Let $r(x) = \min_j d(x, x_j)$. Divide the points into annuli of width h around x . Separation gives $O_{a,d}(m^{d-1})$ points in the m th annulus. Summing (1), and treating the nearest point separately, yields

$$(2) \quad |F_{\text{hi}}(x)| \leq C_{a,d} (h + h^d r(x)^{1-d}).$$

Put $q = d/(d-1)$. If $\lambda > 2C_{\alpha,d}h$, then (2) shows that

$$\{|F_{\text{hi}}| > \lambda\} \subseteq \bigcup_{j=1}^N B(x_j, R_\lambda), \quad R_\lambda = C_{\alpha,d} \left(\frac{h^d}{\lambda} \right)^{1/(d-1)}.$$

Consequently

$$|\{|F_{\text{hi}}| > \lambda\}| \leq C_d N R_\lambda^d \leq C_{\alpha,d} \left(\frac{h}{\lambda} \right)^q.$$

For $\lambda \leq 2C_{\alpha,d}h$ the trivial measure bound gives the same weak-norm conclusion. Hence $\sup_{\lambda>0} \lambda |\{|F_{\text{hi}}| > \lambda\}|^{1/q} \lesssim_{\alpha,d} h$. $\square$

Lemma 2 (The smoothed Kronecker field). *If $x_j = j\alpha \pmod{1}$ and α is badly approximable, there is a vector field F_{lo} such that*

$$\operatorname{div} F_{\text{lo}} = P_t \mu_N - 1 \quad \text{and} \quad \|F_{\text{lo}}\|_2 \leq C_{\alpha,d}h.$$

In particular, $\|F_{\text{lo}}\|_{L^{d/(d-1)},\infty} \leq C_{\alpha,d}h$.

Proof. Take $F_{\text{lo}} = \nabla u$, where $\Delta u = P_t \mu_N - 1$ and u has mean zero. Thus

$$\|F_{\text{lo}}\|_2 = \|P_t \mu_N - 1\|_{\dot{H}^{-1}}.$$

The dyadic Fourier estimate proved in [1, proofs of Theorems 5 and 6] is

$$\|P_t \mu_N - 1\|_{\dot{H}^{-1}} \leq \frac{C_{\alpha,d}}{N t^{(d-1)/2}}.$$

At $t = h^2 = N^{-2/d}$ the right side is $C_{\alpha,d}h$. Finally $d/(d-1) \leq 2$, so on the probability space $\mathbb{T}^d$ the L^2 bound implies the asserted weak-Lorentz bound. $\square$

For completeness, the other input needed by Lemma 1 follows directly from the source definition of bad approximability. If $1 \leq i < j \leq N$, then

$$d(x_i, x_j) \geq \max_{1 \leq \ell \leq d} \|(j-i)\alpha_\ell\|_{\mathbb{R}/\mathbb{Z}} \geq c_\alpha (j-i)^{-1/d} \geq c_\alpha h.$$

FULL ANSWER

Theorem 1. *Let $d \geq 2$ and let $\alpha \in \mathbb{R}^d$ be badly approximable. For $x_j = j\alpha \pmod{1}$ and every differentiable $f : \mathbb{T}^d \rightarrow \mathbb{R}$,*

$$\left| \int_{\mathbb{T}^d} f(x) dx - \frac{1}{N} \sum_{j=1}^N f(x_j) \right| \leq C_{\alpha,d} N^{-1/d} \|\nabla f\|_\infty^{(d-1)/d} \|\nabla f\|_1^{1/d}.$$

The constant is independent of N . Hence both questions following Theorem 7 of [1] have positive answers.

Proof. Let $F = F_{\text{hi}} + F_{\text{lo}}$ be supplied by Lemmas 1 and 2. Then

$$\operatorname{div} F = \mu_N - 1, \quad \|F\|_{L^{d/(d-1)},\infty} \leq C_{\alpha,d}h.$$

Distributional integration by parts and O’Neil’s Lorentz Hölder inequality [2] give

$$\begin{aligned} \left| \int f d(\mu_N - dx) \right| &\leq C_d \|F\|_{L^{d/(d-1),\infty}} \|\nabla f\|_{L^{d,1}} \\ &\leq C_{\alpha,d} h \|\nabla f\|_{L^{d,1}}. \end{aligned}$$

If $M = \|\nabla f\|_\infty$ and $L = \|\nabla f\|_1$, the layer-cake formula and Hölder’s inequality give

$$\begin{aligned} \|\nabla f\|_{L^{d,1}} &= d \int_0^M |\{|\nabla f| > \lambda\}|^{1/d} d\lambda \\ &\leq d M^{(d-1)/d} \left(\int_0^M |\{|\nabla f| > \lambda\}| d\lambda \right)^{1/d} \\ &= d M^{(d-1)/d} L^{1/d}. \end{aligned}$$

Since $h = N^{-1/d}$, this proves the claim. $\square$

VERIFICATION NOTES

The proof does not infer an L^1 endpoint from the source’s W_2 estimate; that would be invalid. It retains the divergence fields before applying duality. The two pieces live in the same critical weak space for different reasons: $L^2 \subset L^{d/(d-1),\infty}$ for the smoothed part, while separation controls the Coulomb singularities of the unsmoothed part.

The powers can be checked from the distribution calculation. A single weighted Coulomb singularity has size $N^{-1}r^{1-d} = h^d r^{1-d}$; N such singularities produce distribution $O((h/\lambda)^{d/(d-1)})$, whose weak norm is exactly $O(h)$.

Smooth functions suffice for the distributional computation. Standard mollification extends the estimate to differentiable/Lipschitz functions with gradient in $L^1 \cap L^\infty$, which is the source’s class.

NOVELTY CHECK AND SCOPE

A bounded local and web search on August 11, 2026 used the arXiv identifier, the exact endpoint sentence, and combinations of “Kronecker sequence”, “ L^1 gradient”, “Lorentz”, “numerical integration”, and “badly approximable”. It found the source and its journal version, later papers on Wasserstein equidistribution, and 2024–2026 work on quasi-uniformity of badly approximable Kronecker sequences. It found no paper claiming this endpoint or using the critical heat-field argument. Novelty is plausible, not certified.

The theorem answers the explicit L^1 question for the same badly approximable vectors and dimensions $d \geq 2$ treated by the source. It does not characterize all sequences satisfying the endpoint, nor decide whether bad approximability is necessary.

HUMAN REVIEW RECOMMENDATION

Check the annular-packing estimate (2) and the sign in the heat-flow divergence split. Then compare Lemma 2 directly with the Fourier sum in the source's proofs of Theorems 5 and 6. If those pass, Theorem 1 solves both stated questions.

REFERENCES

- [1] L. Brown and S. Steinerberger, *On the Wasserstein distance between classical sequences and the Lebesgue measure*, Trans. Amer. Math. Soc. **373** (2020), 8943–8962; arXiv:1909.09046.
- [2] R. O’Neil, *Convolution operators and $L(p, q)$ spaces*, Duke Math. J. **30** (1963), 129–142.